

netCDF

We are going to make an animated gif and not embarrass a friend!

You need to take your time. Every cell is important today. If you rush you will get yourself in trouble.

Next year turn it into a function.

```
In [1]: # The usual libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

%matplotlib ipynb

# The new ones
import netCDF4
import cartopy.crs as ccrs
import cartopy.feature
from cartopy.util import add_cyclic_point

import datetime

import imageio
import os
```

You are going to read in a netcdf file. Plot sea surface temperatures. Animate them for the year and turn in an animated GIF.

The links to get the sea surface data don't always work. Be patient. WE DOWNLOADED THIS YESTERDAY.

- Start here <http://www.esrl.noaa.gov/psd/data/gridded/data.noaa.oisst.v2.highres.html#detail>
- Every year this changes.
- Now I go into the box on the lower left.
- Scroll down to sea surface temperature mean with 45 years. Click on it.
- your file should be named sst.day.mean....
- then click a year and it will download. Do not do this year! It is not a full year because we aren't done yet!
- It is a 400mb file. It might be slow if you are on wifi
- Now make sure it is saved in your working directory and we can open it. url is the file name.
- It is just like the last packet!

```
In [ ]:
```

```
In [2]: file=('sst.day.mean.2024.nc')
f=netCDF4.Dataset(file)
```

You saved the file into "f". So if we print f we will see part of it. But remember the f is not the whole file but more like a function that we can then call to get at the whole file. We will use "f." notation to learn about the file. You know the drill!

```
In [3]: print (f)
```

```
<class 'netCDF4.Dataset'>
root group (NETCDF4_CLASSIC data model, file format HDF5):
  Conventions: CF-1.5
  title: NOAA/NCEI 1/4 Degree Daily Optimum Interpolation Sea Surface Temperature (OISST) Analysis, Version 2.1
  institution: NOAA/National Centers for Environmental Information
  source: NOAA/NCEI https://www.ncei.noaa.gov/data/sea-surface-temperature-optimum-interpolation/v2.1/access/avhrr/
  References: https://www.psl.noaa.gov/data/gridded/data.noaa.oisst.v2.highres.html
  dataset_title: NOAA Daily Optimum Interpolation Sea Surface Temperature
  version: Version 2.1
  comment: Reynolds, et al.(2007) Daily High-Resolution-Blended Analyses for Sea Surface Temperature (available at https://doi.org/10.1175/2007JCLI1824.1). Banzon, et al.(2016) A long-term record of blended satellite and in situ sea-surface temperature for climate monitoring, modeling and environmental studies (available at https://doi.org/10.5194/essd-8-165-2016). Huang, B., C. Liu, V. Banzon, E. Freeman, G. Graham, B. Hankins, T. Smith, and H.-M. Zhang, 2021: Improvements of the Daily Optimum Interpolation Sea Surface Temperature (DOISST) Version 2.1 (available at https://doi.org/10.1175/JCLI-D-20-0166.1). Anomaly is based on 1971-2000 OI.v2 SST. Satellite data: multi-satellite sensor (AVHRR+VIIRS) data. Ice data: NCEP Ice and GSFC Ice. Data less than 15 days old may be subject to revision.
  dimensions(sizes): time(366), lat(720), lon(1440)
  variables(dimensions): float64 time(time), float32 lat(lat), float32 lon(lon), float32 sst(time, lat, lon)
  groups:
```

So what do we have? We have 366 time steps of lat and long with sea surface temperatures at every day. so lets try to pull it out. It is easy....

```
In [4]: print (f.variables['time'])
```

```
<class 'netCDF4.Variable'>
float64 time(time)
  long_name: Time
  units: days since 1800-01-01 00:00:00
  delta_t: 0000-00-01 00:00:00
  avg_period: 0000-00-01 00:00:00
  axis: T
  actual_range: [81814. 82179.]
unlimited dimensions: time
current shape = (366,)
filling on, default _FillValue of 9.969209968386869e+36 used
```

We are going to come back to the time values.

Now lets look at latitude.

```
In [5]: print (f.variables['lat'])
```

```
<class 'netCDF4.Variable'>
float32 lat(lat)
  long_name: Latitude
  standard_name: latitude
  units: degrees_north
  actual_range: [-89.875  89.875]
  axis: Y
unlimited dimensions:
current shape = (720,)
filling on, default _FillValue of 9.969209968386869e+36 used
```

There are 720 locations for latitude. Or 4 values every degree

Now lets look at longitude

```
In [6]: print (f.variables['lon'])
```

```

<class 'netCDF4.Variable'>
float32 lon(lon)
  long_name: Longitude
  standard_name: longitude
  units: degrees_east
  actual_range: [1.25000e-01 3.59875e+02]
  axis: X
unlimited dimensions:
current shape = (1440,)
filling on, default _FillValue of 9.969209968386869e+36 used

```

longitude has 1440 values or twice as many as latitude as it goes around the whole globe

Now we can look at the actual sea surface temperature.

```
In [7]: print (f.variables['sst'])
```

```

<class 'netCDF4.Variable'>
float32 sst(time, lat, lon)
  long_name: Daily Sea Surface Temperature
  units: degC
  valid_range: [-3. 45.]
  missing_value: -9.96921e+36
  precision: 2.0
  dataset: NOAA High-resolution Blended Analysis
  var_desc: Sea Surface Temperature
  level_desc: Surface
  statistic: Mean
  parent_stat: Individual Observations
  actual_range: [-1.8 36.13]
unlimited dimensions: time
current shape = (366, 720, 1440)
filling on, default _FillValue of 9.969209968386869e+36 used

```

The sst is really cool. Think of it as 366 days of data, each in an array of values in the shape of 720 by 1440. So for each day we will have a map of sea surface temperatures.

Can we actually look at some numbers?

```
In [8]: f.variables['lat'][:]
```

```
Out[8]: masked_array(data=[-89.875, -89.625, -89.375, -89.125, -88.875, -88.625,
                             -88.375, -88.125, -87.875, -87.625, -87.375, -87.125,
                             -86.875, -86.625, -86.375, -86.125, -85.875, -85.625,
                             -85.375, -85.125, -84.875, -84.625, -84.375, -84.125,
                             -83.875, -83.625, -83.375, -83.125, -82.875, -82.625,
                             -82.375, -82.125, -81.875, -81.625, -81.375, -81.125,
                             -80.875, -80.625, -80.375, -80.125, -79.875, -79.625,
                             -79.375, -79.125, -78.875, -78.625, -78.375, -78.125,
                             -77.875, -77.625, -77.375, -77.125, -76.875, -76.625,
                             -76.375, -76.125, -75.875, -75.625, -75.375, -75.125,
                             -74.875, -74.625, -74.375, -74.125, -73.875, -73.625,
                             -73.375, -73.125, -72.875, -72.625, -72.375, -72.125,
                             -71.875, -71.625, -71.375, -71.125, -70.875, -70.625,
                             -70.375, -70.125, -69.875, -69.625, -69.375, -69.125,
                             -68.875, -68.625, -68.375, -68.125, -67.875, -67.625,
                             -67.375, -67.125, -66.875, -66.625, -66.375, -66.125,
                             -65.875, -65.625, -65.375, -65.125, -64.875, -64.625,
                             -64.375, -64.125, -63.875, -63.625, -63.375, -63.125,
                             -62.875, -62.625, -62.375, -62.125, -61.875, -61.625,
                             -61.375, -61.125, -60.875, -60.625, -60.375, -60.125,
                             -59.875, -59.625, -59.375, -59.125, -58.875, -58.625,
                             -58.375, -58.125, -57.875, -57.625, -57.375, -57.125,
                             -56.875, -56.625, -56.375, -56.125, -55.875, -55.625,
                             -55.375, -55.125, -54.875, -54.625, -54.375, -54.125,
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                             -50.875, -50.625, -50.375, -50.125, -49.875, -49.625,
                             -49.375, -49.125, -48.875, -48.625, -48.375, -48.125,
                             -47.875, -47.625, -47.375, -47.125, -46.875, -46.625,
                             -46.375, -46.125, -45.875, -45.625, -45.375, -45.125,
                             -44.875, -44.625, -44.375, -44.125, -43.875, -43.625,
                             -43.375, -43.125, -42.875, -42.625, -42.375, -42.125,
                             -41.875, -41.625, -41.375, -41.125, -40.875, -40.625,
                             -40.375, -40.125, -39.875, -39.625, -39.375, -39.125,
                             -38.875, -38.625, -38.375, -38.125, -37.875, -37.625,
                             -37.375, -37.125, -36.875, -36.625, -36.375, -36.125,
                             -35.875, -35.625, -35.375, -35.125, -34.875, -34.625,
                             -34.375, -34.125, -33.875, -33.625, -33.375, -33.125,
                             -32.875, -32.625, -32.375, -32.125, -31.875, -31.625,
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                             -28.375, -28.125, -27.875, -27.625, -27.375, -27.125,
                             -26.875, -26.625, -26.375, -26.125, -25.875, -25.625,
                             -25.375, -25.125, -24.875, -24.625, -24.375, -24.125,
                             -23.875, -23.625, -23.375, -23.125, -22.875, -22.625,
                             -22.375, -22.125, -21.875, -21.625, -21.375, -21.125,
                             -20.875, -20.625, -20.375, -20.125, -19.875, -19.625,
                             -19.375, -19.125, -18.875, -18.625, -18.375, -18.125,
                             -17.875, -17.625, -17.375, -17.125, -16.875, -16.625,
                             -16.375, -16.125, -15.875, -15.625, -15.375, -15.125,
                             -14.875, -14.625, -14.375, -14.125, -13.875, -13.625,
                             -13.375, -13.125, -12.875, -12.625, -12.375, -12.125,
                             -11.875, -11.625, -11.375, -11.125, -10.875, -10.625,
                             -10.375, -10.125, -9.875, -9.625, -9.375, -9.125,
                             -8.875, -8.625, -8.375, -8.125, -7.875, -7.625,
                             -7.375, -7.125, -6.875, -6.625, -6.375, -6.125,
                             -5.875, -5.625, -5.375, -5.125, -4.875, -4.625,
                             -4.375, -4.125, -3.875, -3.625, -3.375, -3.125,
                             -2.875, -2.625, -2.375, -2.125, -1.875, -1.625,
                             -1.375, -1.125, -0.875, -0.625, -0.375, -0.125,
                             0.125, 0.375, 0.625, 0.875, 1.125, 1.375,
                             1.625, 1.875, 2.125, 2.375, 2.625, 2.875,
                             3.125, 3.375, 3.625, 3.875, 4.125, 4.375,
                             4.625, 4.875, 5.125, 5.375, 5.625, 5.875,
                             6.125, 6.375, 6.625, 6.875, 7.125, 7.375,
                             7.625, 7.875, 8.125, 8.375, 8.625, 8.875,
                             9.125, 9.375, 9.625, 9.875, 10.125, 10.375,
                             10.625, 10.875, 11.125, 11.375, 11.625, 11.875,
```

```

12.125, 12.375, 12.625, 12.875, 13.125, 13.375,
13.625, 13.875, 14.125, 14.375, 14.625, 14.875,
15.125, 15.375, 15.625, 15.875, 16.125, 16.375,
16.625, 16.875, 17.125, 17.375, 17.625, 17.875,
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19.625, 19.875, 20.125, 20.375, 20.625, 20.875,
21.125, 21.375, 21.625, 21.875, 22.125, 22.375,
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42.125, 42.375, 42.625, 42.875, 43.125, 43.375,
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46.625, 46.875, 47.125, 47.375, 47.625, 47.875,
48.125, 48.375, 48.625, 48.875, 49.125, 49.375,
49.625, 49.875, 50.125, 50.375, 50.625, 50.875,
51.125, 51.375, 51.625, 51.875, 52.125, 52.375,
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85.625, 85.875, 86.125, 86.375, 86.625, 86.875,
87.125, 87.375, 87.625, 87.875, 88.125, 88.375,
88.625, 88.875, 89.125, 89.375, 89.625, 89.875],
    mask=False,
    fill_value=1e+20,
    dtype=float32)

```

So we can pull out numbers and look at them. They all come out in numpy arrays.

```
In [9]: type(f.variables['lat'][:])
```

```
Out[9]: numpy.ma.core.MaskedArray
```

But we can pull them out and set them to a variable. We are not going to use Pandas and just leave everything in numpy arrays for today. This might take a minute because you are putting 400 Mb of data into memory. Your computer might get cranky and we might have to restart at some point.

```
In [10]: lon=f.variables['lon'][:]  
lat=f.variables['lat'][:]  
sst=f.variables['sst'][:]  
time=f.variables['time'][:]
```

You can look at the shape of each element. Remember 1 year has 366 days! 2024 was a Leap year!

```
In [11]: np.shape(time)
```

```
Out[11]: (366,)
```

```
In [12]: np.shape(lon)
```

```
Out[12]: (1440,)
```

```
In [13]: np.shape(lat)
```

```
Out[13]: (720,)
```

```
In [14]: np.shape(sst)
```

```
Out[14]: (366, 720, 1440)
```

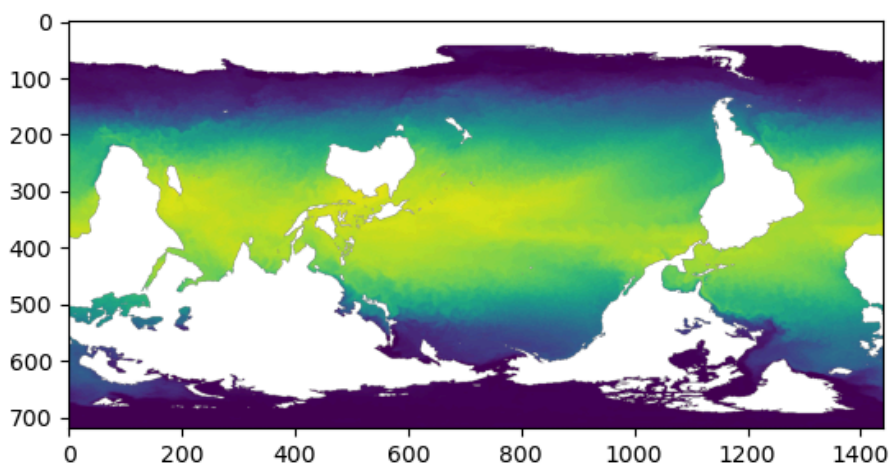
So what do we have? We have a lat/lon grid. Then each point has a sst. Then we have that repeated 365 times for each day of the year.

Let's make our easy plot and then we will make the nice map.

```
In [15]: fig,ax=plt.subplots()  
ax.imshow(sst[0])
```

```
Out[15]: <matplotlib.image.AxesImage at 0x138174cd0>
```

Figure



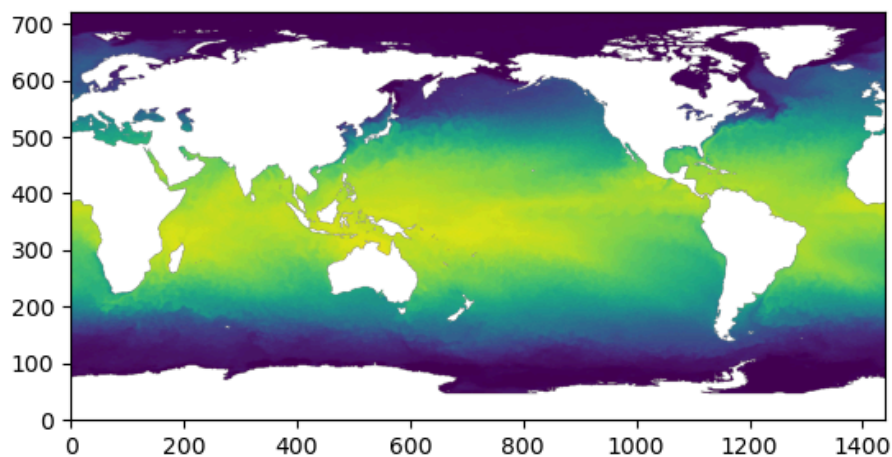
What just happened? This did not make a map? It just took the sst array and plotted it. Each array point has a latitude and longitude we could use to make a map. But we didn't do that. We just did a raw imshow which shows the array. It is a nice start. Read the help and see if you can flip the array?

```
In [16]: #?plt.imshow
```

```
In [17]: fig,ax=plt.subplots()  
ax.imshow(sst[0],origin='lower')
```

```
Out[17]: <matplotlib.image.AxesImage at 0x1381af650>
```

Figure

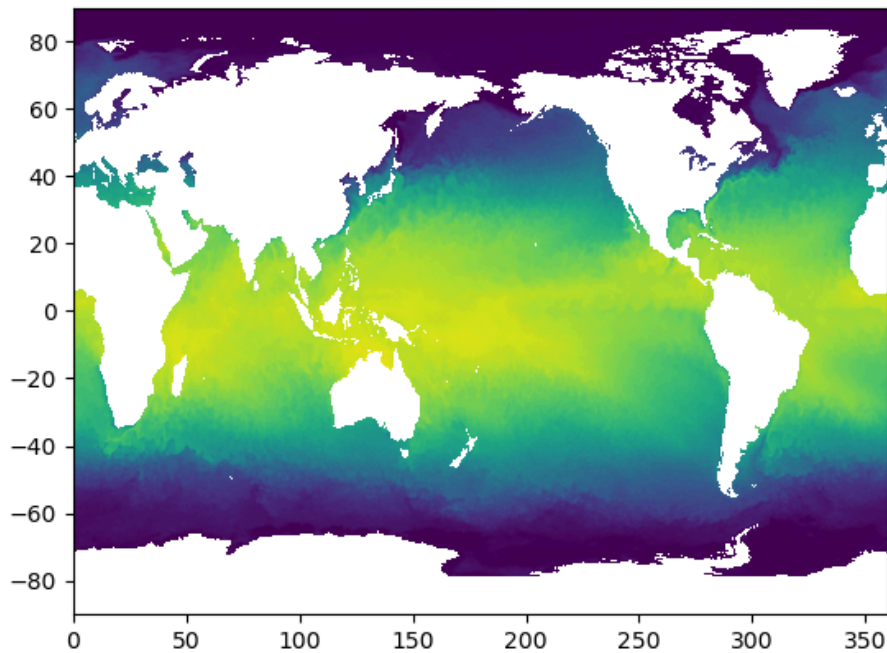


imshow is nice as a quick way to show an array but it is not a map. The next way to make a better map is pcolormesh. It uses the lat and long so it is better but still not a map.

```
In [18]: fig,ax=plt.subplots()  
ax.pcolormesh(lon,lat,sst[0])
```

```
Out[18]: <matplotlib.collections.QuadMesh at 0x138203e90>
```

Figure



Lets try to make that nicer

- Let's go back and grab our last map and paste it in here. You should be able to make it work

```
In [19]: fig,ax=plt.subplots(subplot_kw={"projection":ccrs.PlateCarree(central_longitude=-120)})
fig.set_size_inches(7.5,5)

ax.coastlines()
ax.add_feature(cartopy.feature.BORDERS)
ax.gridlines(draw_labels=True) #see if you like them.

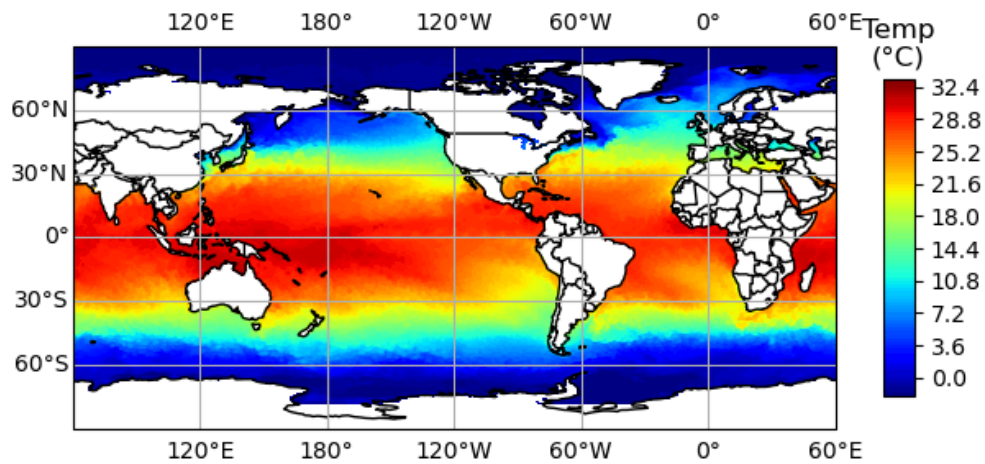
data, lonW = add_cyclic_point(sst[0], coord=lon)

air_contour=ax.contourf(lonW, lat, data,
                        transform=ccrs.PlateCarree(),
                        cmap='jet',levels=100)

cbar_ax=fig.colorbar(air_contour,ax=ax,shrink=0.5, aspect=20*0.5) #shrink is the size and aspect
cbar_ax.ax.set_title('Temp\n(\N{DEGREE SIGN}C)')

Out[19]: Text(0.5, 1.0, 'Temp\n(°C)')
```


Figure



Let's make it look better

- I am going to make the land a slightly different color
- Then we need to set the levels better. The reason is different days have different ranges.
- We will use linspace to get levels and then pass the levels. You are setting the range. For contourf you need to be over the min and max.

```
In [20]: fig,ax=plt.subplots(subplot_kw={"projection":ccrs.PlateCarree(central_longitude=-120)})
fig.set_size_inches(7.5,5)

ax.coastlines()
ax.add_feature(cartopy.feature.BORDERS,edgecolor='k')
ax.add_feature(cartopy.feature.LAND,facecolor='xkcd:off white')
ax.gridlines(draw_labels=True) #see if you like them.

data, lonW = add_cyclic_point(sst[0], coord=lon)

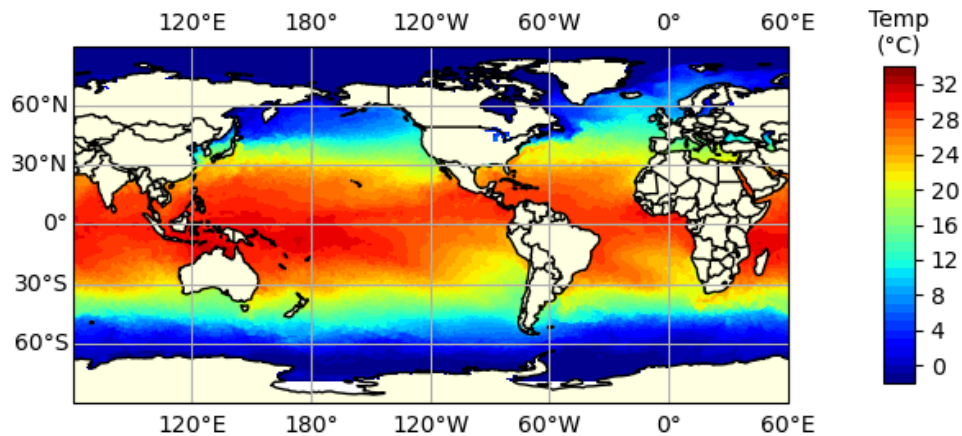
levels=np.linspace(-2,34,37)

air_contour=ax.contourf(lonW, lat, data,
                        transform=ccrs.PlateCarree(),
                        cmap='jet',levels=levels)

cbar_ax=fig.colorbar(air_contour,ax=ax,shrink=0.5, aspect=20*0.5,pad=0.1) #shrink is the size an
cbar_ax.ax.set_title('Temp\n(N{DEGREE SIGN}C)',fontsize=10)

Out[20]: Text(0.5, 1.0, 'Temp\n(°C)')
```

Figure



remember that we have 366 days of data.

- So lets set a parameter to day number that we can change
- If you know me you should know a for loop is coming and we are stepping our way there.
- Now make a variable called "iday" and set it equal to the day you want to plot. For example January 31 will be 30 because python starts at 0.
- Compare day 0 and day 180. Do the temperatures look different?

```
In [21]: iday=1

fig,ax=plt.subplots(subplot_kw={"projection":ccrs.PlateCarree(central_longitude=-120)})
fig.set_size_inches(7.5,5)

ax.coastlines()
ax.add_feature(cartopy.feature.BORDERS,edgecolor='k')
ax.add_feature(cartopy.feature.LAND,facecolor='xkcd:off white')
ax.gridlines(draw_labels=True) #see if you like them.

data, lonW = add_cyclic_point(sst[iday], coord=lon)

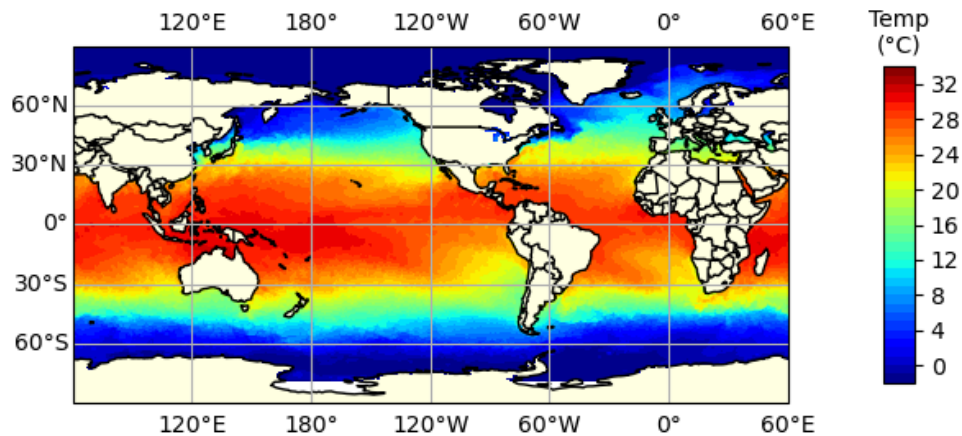
levels=np.linspace(-2,34,37)

air_contour=ax.contourf(lonW, lat, data,
                        transform=ccrs.PlateCarree(),
                        cmap='jet',levels=levels)

cbar_ax=fig.colorbar(air_contour,ax=ax,shrink=0.5, aspect=20*0.5,pad=0.1) #shrink is the size an
cbar_ax.ax.set_title('Temp\n(\N{DEGREE SIGN}C)',fontsize=10)
```

```
Out[21]: Text(0.5, 1.0, 'Temp\n(°C)')
```

Figure



That is a good looking map!!!!

I am starting to see a path forward for how to make a movie of the temperatures over the year.

- Python is not perfect for making movies.
- We are going to make a modern flip book.
- We will save an image for each frame and then we can use python or a web program to put them together. It is like making a burst on your phone into a movie.
- We want to end with an animated GIF with python or a program. Things we will need to be able to do.

How do we do thi??

- Loop over all the data.
- Figure out the date for the data.
- plot the data.
- save the data in a unique file for each day
- use python/website to put it all together.

So we need to learn each of these steps. I will start with figuring out the date for each slice.

Back to our crazy dates!

```
In [26]: print (f.variables['time'])
```

```
<class 'netCDF4._netCDF4.Variable'>
float64 time(time)
  long_name: Time
  units: days since 1800-01-01 00:00:00
  delta_t: 0000-00-01 00:00:00
  avg_period: 0000-00-01 00:00:00
  axis: T
  actual_range: [81814. 82179.]
unlimited dimensions: time
current shape = (366,)
filling on, default _FillValue of 9.969209968386869e+36 used
```

Still crazy!

Time since 1800 in days! Let's deal with it like last time. Go cut and paste what we did.

- Set your startdate
- calculate your datedelta
- Add them together to figure out your date
- use ndays again
- Change ndays and see if it

```
In [22]: iday=180
startdate=datetime.datetime(1800,1,1)
datedelta=datetime.timedelta(days=time[iday])
mapdate=startdate+datedelta
print ("The date is {:%B %d, %Y}".format(mapdate))
```

The date is June 29, 2024

We need a for loop

- I do not know the best way to for loop here.
- I see three options
- just give us a number to loop 20. e.g. 365. But would get an error on a leap year.
- We could do an enumerate loop.
- We could also just loop to length of a parameter.
- I am going to do np.arange(len(time))
- then print iday
- Important note. I added [0:10] I did this to save paper and only print 10. You can delete it.

```
In [24]: for iday in np.arange(len(time)):
        #print(iday) #uncomment. Trying to save paper
```

```
Cell In[24], line 2
    #print(iday) #uncomment. Trying to save paper
    ^
```

SyntaxError: incomplete input

If you only wanted to loop over the first 10 while figuring things out do this

```
In [25]: for iday in np.arange(len(time))[0:10]:
        print(iday) #uncomment. Trying to save paper
```

```
0
1
2
3
4
5
6
7
8
9
```

Now print each day in the for loop. to save memory put the least amount of things needed in a for loop. You can do this!

```
In [26]: startdate=datetime.datetime(1800,1,1)

for iday in np.arange(len(time))[0:10]:
    datedelta=datetime.timedelta(days=time[iday])
    mapdate=startdate+datedelta
    print ("The date is {:%B %d, %Y}".format(mapdate))
```

```
The date is January 01, 2024
The date is January 02, 2024
The date is January 03, 2024
The date is January 04, 2024
The date is January 05, 2024
The date is January 06, 2024
The date is January 07, 2024
The date is January 08, 2024
The date is January 09, 2024
The date is January 10, 2024
```

Now make a loop where you give the count and the day. So day 0 is January 1.

```
In [27]: startdate=datetime.datetime(1800,1,1)
for iday in np.arange(len(time))[0:10]:
    datedelta=datetime.timedelta(days=time[iday])
    mapdate=startdate+datedelta
    print ("The date on day {} is {:%B %d, %Y}".format(iday,mapdate))
```

```
The date on day 0 is January 01, 2024
The date on day 1 is January 02, 2024
The date on day 2 is January 03, 2024
The date on day 3 is January 04, 2024
The date on day 4 is January 05, 2024
The date on day 5 is January 06, 2024
The date on day 6 is January 07, 2024
The date on day 7 is January 08, 2024
The date on day 8 is January 09, 2024
The date on day 9 is January 10, 2024
```

this might not make sense.

If we are going to make 366 plots one for each day. We need to save each to a file. So we need to create a file name we can send to savefig. You can't type 365 file names. We need to automate!

- We will want our files nicely organized.
- We could use the dates to name the files. But we will lose their order. You could name the file yyyyymmdd and that should keep them in order. I haven't tried yet.
- Instead I will do a number in the file name.
- But we will want a three digit number in the name. This will keep them all in order.
- it will look like
 - sst_000.png
 - sst_001.png
 - sst_002.png
 - sst_365.png.
- We can use format to pad the numbers. You remember this?

<https://stackoverflow.com/questions/339007/nicest-way-to-pad-zeroes-to-string>

```
In [28]: startdate=datetime.datetime(1800,1,1)
for iday in np.arange(len(time))[0:10]:
    datedelta=datetime.timedelta(days=time[iday])
    mapdate=startdate+datedelta
```

```
filename="sst_{:03d}.png".format(iday)
print ("The date on day {} is {:%B %d, %Y} with filename {}".format(iday,mapdate,filename))
```

```
The date on day 0 is January 01, 2024 with filename sst_000.png
The date on day 1 is January 02, 2024 with filename sst_001.png
The date on day 2 is January 03, 2024 with filename sst_002.png
The date on day 3 is January 04, 2024 with filename sst_003.png
The date on day 4 is January 05, 2024 with filename sst_004.png
The date on day 5 is January 06, 2024 with filename sst_005.png
The date on day 6 is January 07, 2024 with filename sst_006.png
The date on day 7 is January 08, 2024 with filename sst_007.png
The date on day 8 is January 09, 2024 with filename sst_008.png
The date on day 9 is January 10, 2024 with filename sst_009.png
```

This confuses people!!!!

- I will want to put the files into a subfolder.
- 366 files in your main folder is a mess!
- So make a folder named "sst" where you are working on your computer. Go make it
- Now we can trick python with the file name
- we add the sst/ prefix to the name.
- the computer is tricked into putting the file into that folder
- I am going to repeat this. You named it sst/ but the folder doesn't exist until you make it and then it will put the files into it.

```
In [29]: startdate=datetime.datetime(1800,1,1)
for iday in np.arange(len(time))[0:10]:
    datedelta=datetime.timedelta(days=time[iday])
    mapdate=startdate+datedelta
    filename="sst/sst_{:03d}.png".format(iday)
    print ("The date on day {} is {:%B %d, %Y} with filename {}".format(iday,mapdate,filename))
```

```
The date on day 0 is January 01, 2024 with filename sst/sst_000.png
The date on day 1 is January 02, 2024 with filename sst/sst_001.png
The date on day 2 is January 03, 2024 with filename sst/sst_002.png
The date on day 3 is January 04, 2024 with filename sst/sst_003.png
The date on day 4 is January 05, 2024 with filename sst/sst_004.png
The date on day 5 is January 06, 2024 with filename sst/sst_005.png
The date on day 6 is January 07, 2024 with filename sst/sst_006.png
The date on day 7 is January 08, 2024 with filename sst/sst_007.png
The date on day 8 is January 09, 2024 with filename sst/sst_008.png
The date on day 9 is January 10, 2024 with filename sst/sst_009.png
```

Now we know how to

- Make a title name with the date.
- make a file name
- loop over the array

But before you begin you need to think a little.

You need to think!!!!!! Else your computer will crash

Each timestep we are going to make a figure, make a map, and save it.

But making a figure and map are "expensive" and eat up cpu. And you need to do this 365 times.

But unfortunately we can't use clear axis. It doesn't work with cartopy. So you need to open and CLOSE a figure during each step. You need to close the figure each time. Else your computer will crash. you close the figure by doing `plt.close(fig)`

So i would test things by running just a few days. use your `[0:10]` Plus you can use `kernal->interrupt` to stop a run.

Don't forget a title with your name and date. You need your name in the title to get credit for the homework. Otherwise a zero.

Repeating. you need a title with your name and date

TRICK: you are making 365 files. make them small. use dpi=50. So you call to savefig will look like `fig.savefig(filename,dpi=50,bbox_inches='tight',facecolor='white')` You need to add the `facecolor='white'` to make the titles look good.

dpi=50 is probably too small. If you have space on your computer I would use dpi=100. The map comes out much nicer.

The websites for making a gif are full or viruses. I would use python.

I found this stackoverflow example.

<https://stackoverflow.com/questions/41228209/making-gif-from-images-using-imageio-in-python>

Look at the answer below and just update with your directories!

BUT THERE IS A BIG PROBLEM! A few dates are missing. How to fix

- The cheating way is to figuring out the missing ones and copy the day before or after into it.
- Learn how to use try and except. This is a great python trick.
 - try is what you want to try. But if it fails it won't crash the program
 - except is what happens when there is a warning.
 - So look it up. Try your code and add an except.

PS. This is slow. My computer takes 30+ minutes and makes a lot of noise!

PPS. You cannot open your movie in preview. Use a web browser.

PPPS. I repeat. open in a web browser no preview.

HW Final 10%

- Get rid of your weird days.
- You can delete and then use try/except
- You can delete and repeat the previous day with a new file name.

In [37]:

```
The date on day 0 is January 01, 2024 with filename sst/sst_000.png
The date on day 1 is January 02, 2024 with filename sst/sst_001.png
The date on day 2 is January 03, 2024 with filename sst/sst_002.png
The date on day 3 is January 04, 2024 with filename sst/sst_003.png
The date on day 4 is January 05, 2024 with filename sst/sst_004.png
The date on day 5 is January 06, 2024 with filename sst/sst_005.png
The date on day 6 is January 07, 2024 with filename sst/sst_006.png
The date on day 7 is January 08, 2024 with filename sst/sst_007.png
The date on day 8 is January 09, 2024 with filename sst/sst_008.png
The date on day 9 is January 10, 2024 with filename sst/sst_009.png
```

Make the GIF. Choose a method.....

In []:

In [32]: `### option 1.`

```
images = []
for i in range(0,366):
```

```

filename='sst/sst_{:03d}.png'.format(i)
images.append(imageio.v2.imread(filename))

#imageio.mimwrite('sst/omg_movie2.gif', images)

imageio.v2.mimsave('sst/omg_movie2.gif', images)
#imageio.mimsave('sst/omg_movie2.mp4', images)

```

something wrong on step 28
something wrong on step 90
something wrong on step 203
something wrong on step 278
something wrong on step 306

```

In [ ]: ### option 2
png_dir = 'sst/'
images = []
for file_name in sorted(os.listdir(png_dir)):
    if file_name.endswith('.png'):
        file_path = os.path.join(png_dir, file_name)
        images.append(imageio.imread(file_path))

imageio.mimsave('sst/omg_movie2.gif', images)

```

In []:

ANSWERS

```

In [34]: startdate=datetime.datetime(1800,1,1)

for iday in np.arange(len(time))[0:10]:
    datedelta=datetime.timedelta(days=time[iday])
    mapdate=startdate+datedelta
    print ("The date is {:%B %d, %Y}".format(mapdate))

```

The date is January 01, 2024
The date is January 02, 2024
The date is January 03, 2024
The date is January 04, 2024
The date is January 05, 2024
The date is January 06, 2024
The date is January 07, 2024
The date is January 08, 2024
The date is January 09, 2024
The date is January 10, 2024

```

In [37]: startdate=datetime.datetime(1800,1,1)
for iday in np.arange(len(time))[0:10]:
    datedelta=datetime.timedelta(days=time[iday])
    mapdate=startdate+datedelta
    print ("The date on day {} is {:%B %d, %Y}".format(iday,mapdate))

```

The date on day 0 is January 01, 2024
The date on day 1 is January 02, 2024
The date on day 2 is January 03, 2024
The date on day 3 is January 04, 2024
The date on day 4 is January 05, 2024
The date on day 5 is January 06, 2024
The date on day 6 is January 07, 2024
The date on day 7 is January 08, 2024
The date on day 8 is January 09, 2024
The date on day 9 is January 10, 2024

Some extra stuff for me.

<https://stackoverflow.com/questions/59014789/making-a-clean-high-quality-gif-with-fix-colorbar-for-multiple-xarray-plots>

In [89]: `!pip install imageio-ffmpeg`

```
Collecting imageio-ffmpeg
  Downloading imageio_ffmpeg-0.4.5-py3-none-macosx_10_9_intel.macosx_10_9_x86_64.macosx_10_10_intel.macosx_10_10_x86_64.whl (22.5 MB)
    |████████████████████████████████████████| 22.5 MB 18.0 MB/s eta 0:00:01
    |██████████| 3.9 MB 6.5 MB/s eta 0:00:03 |██████████| 11.4 MB 6.5 MB/s eta 0:00:02
    |██████████| 15.0 MB 6.5 MB/s eta 0:00:02 |██████████|
    |██████████| 21.2 MB 18.0 MB/s eta 0:00:01
Installing collected packages: imageio-ffmpeg
Successfully installed imageio-ffmpeg-0.4.5
```

In [81]: `?imageio.mimsave`

In []:

In [87]: `from PIL import Image`

```
In [ ]: from PIL import Image
img, *imgs = [Image.open(f) for f in fns_image]
img.save(fp=save_gif_dir+gif_name+'1.gif', format='GIF', append_images=imgs,
        save_all=True, duration=200, loop=0)
```